ISMAEL ESPINOZA JARAMILLO

Robot Learning Researcher | Bimanual Dexterous Manipulation | Imitation + RL | Sim-to-Real Deployment

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Gyeonggi-do, South Korea | Google Scholar: scholar.google.com/citations?user=3S9BoVAAAAAJ

DOB: 1997-03-19 | Visa: D-2 (valid to Mar 2027); E-7-1 on hiring; K-STAR (F-2-7S) in preparation

Summary

Robot learning researcher (Ph.D. expected Feb 2027, Kyung Hee University) working on sim-to-real reinforcement learning, imitation learning, and visuomotor distillation for 44-DoF bimanual dexterous manipulation. Policies trained in an Isaac Lab digital twin completed four long-horizon tasks on table-mounted and upper-humanoid hardware in 18–30 of 35 trials, and four tasks from two RGB cameras alone at 72–84% success. 4 journal articles (incl. IEEE RA-L, presented at IROS 2025), 8 international conference papers (ICRA, HRI, EMBC, UR, ICCR) and 1 symposium paper; 1 granted patent and 2 pending.

Experience

Graduate Researcher (Ph.D.), Robot Learning Mar 2022 – Present

Kyung Hee University, Yongin, South Korea | Advisor: Prof. Tae-Seong Kim

- Engineered a Real2Sim2Real system from glove teleoperation to ROS 2 deployment with FoundationPose 6D tracking, completing four long-horizon tasks on table-mounted and upper-humanoid hardware in 18–30 of 35 trials (behavior cloning alone: 2–13).
- Owned real-robot deployment of the 44-DoF policies: ROS 2 bridge (policy 10 Hz to arm drivers at 125 Hz and hand drivers at 100 Hz), action filter, pose and velocity clamps, and a joint-limit gate; classified all 36 failures in 140 real-robot trials by cause (grasp slips, occlusion, handover misalignment, placement, time limit).
- Co-developed TMAD online teacher-student distillation (second author; patent pending), reaching 72–84% real-robot success on four tasks from two RGB cameras alone (no pose estimator or object model), 44–60 points above a single-agent DextrAH-RGB-style student on three of the four tasks.
- Built an Isaac Lab drive-gain calibration and domain randomization pipeline with up to 500 parallel GPU environments, limiting the success drop from digital twin to real robot to 5–11 points (behavior cloning alone: up to 23).
- Built a glove teleoperation rig (Manus, VIVE, GeoRT retargeting) and a pipeline that recombines subtask segments and keeps only successful replays, training each hardware task from 30 teleoperated demonstrations (300 after augmentation).
- Co-developed TC-TRL (IEEE RA-L 2024, presented at IROS 2025; co-first author; granted patent), a temporal-context multi-agent transformer trained by behavior cloning, then PPO, raising mean success on five simulated long-horizon tasks to 89.9% (best baselines: 59.0% transformer, 26.6% PPO).
- Earlier: recognized 8 activities in real time at 97.35% accuracy on a Jetson Nano from exoskeleton IMU and encoder signals (Sensors 2022, co-first author), and forecast activity with sequence-to-sequence networks (Sensors 2023, first author); 120+ citations combined.
- Headed the hardware deployment pipeline across all projects, managing ROS 2 drivers and real-time state-action telemetry. Kept real-world execution of the trained RL and visuomotor policies on the 44-DoF setup stable through safety middleware code that clamps out-of-bounds joint positions, restricts unsafe velocities, and smooths high-frequency flickering actions.

Research Assistant, Mobile Robotics Mar 2021 – Feb 2022

Universidad de las Fuerzas Armadas (ESPE), Quito, Ecuador

- Delivered a social-navigation and telepresence mobile robot that runs deep-learning perception and navigation in real time on board, by building its ROS software stack (Python, C++), BLE indoor localization, low-level electronics, and PID wheel control.

Pre-professional Intern, Automation Projects Sep 2020 – Oct 2020

SEIUS S.A., Quito, Ecuador

- Drew first-draft AutoCAD Electrical designs of control panels for industrial plants: connection diagrams, protection layers, and the selection and sizing of PLCs, relays, sensors and protection devices.

Education

Ph.D. (integrated M.S.–Ph.D.), Electronics & Information Convergence Eng. Mar 2022 – Feb 2027 (exp.)

Kyung Hee University, South Korea | Major: Biomedical Engineering | GPA 4.27/4.30

Dissertation: A Real2Sim2Real System for Bimanual Long-Horizon Dexterous Manipulation via Temporal-Context Transformer Reinforcement Learning

B.Sc., Electronics, Automation and Control Engineering Sep 2015 – Mar 2021

Universidad de las Fuerzas Armadas (ESPE), Ecuador | GPA 17.15/20.0

Honors & Funding

- President's Scholarship, Kyung Hee University, all 6 semesters (2022–2024)
- BK21 FOUR Scholarship (2023–2024) and Fellowship (2025)
- Journal Publication Scholarship, Kyung Hee University (2023)

Selected Publications

J.-H. Oh*, I. Espinoza*, D. Jung, T.-S. Kim. “Bimanual long-horizon manipulation via temporal-context transformer RL.” IEEE Robotics and Automation Letters (RA-L), 9(12), 10898–10905, 2024. Presented at IROS 2025. (*equal contribution)

J.-H. Oh, D. Jung, I. Espinoza, et al. “Multi-modal affordance planner with temporal-context action policy for long-horizon bimanual robot manipulation.” IEEE International Conference on Robotics and Automation (ICRA), pp. 637–644, 2026.

J.-H. Oh*, I. Espinoza*, et al. “Bimanual long-horizon lifecare robotics with temporal context LLM planner and transformer RL.” IEEE EMBC, 2025. (*equal contribution)

I. Espinoza, Y.-H. Choi, K. Lee, J.-H. Oh, K. J. Pahk, T.-S. Kim. “A Real2Sim2Real transformer policy system for bimanual R2R2H tasks.” ACM/IEEE HRI 2026 Companion, pp. 716–720.

Y.-H. Choi, I. Espinoza, K. Lee, S. Park, J. Y. Kim, J.-H. Oh, W. H. Lee, T.-S. Kim. “A semi-humanoid Real2Sim2Real system for bimanual long-horizon manipulation.” Int. Conf. on Ubiquitous Robots (UR), 2026, in press.

S. Park, I. Espinoza, G. Kim, J. Y. Kim, K. Lee, T.-S. Kim. “TMAD: Temporal multi-agent distillation for bimanual long-horizon manipulation.” International Conference on Control and Robotics (ICCR), 2026, accepted.

I. Espinoza Jaramillo*, J. G. Jeong*, et al. “Real-time human activity recognition with IMU and encoder sensors in wearable exoskeleton robot via deep learning networks.” Sensors, 22(24), 9690, 2022. (*equal contribution)

I. Espinoza Jaramillo, C. Chola, J. G. Jeong, J.-H. Oh, H. Jung, J. H. Lee, W. H. Lee, T.-S. Kim. “Human activity prediction based on forecasted IMU activity signals by sequence-to-sequence deep neural networks.” Sensors, 23(14), 6491, 2023.

Also 6 domestic peer-reviewed conference papers (KICS, 2022–2024); full list on Google Scholar.

Patents

T.-S. Kim, I. Espinoza Jaramillo, J.-H. Oh. System and method for long-horizon task execution of a dual-arm robot using temporal-context transformer network-based reinforcement learning. Korean Patent 10-2943203, granted Mar 2026.

T.-S. Kim, S.-H. Park, G.-B. Kim, I. Espinoza Jaramillo. Vision-based learning method for long-horizon bimanual robot manipulation via temporal multi-agent distillation, and robot system using the same. Korean patent application pending, 2026.

J.-H. Oh, I. Espinoza Jaramillo, T.-S. Kim. Transformer reinforcement learning system with temporal context planning for bimanual long-horizon tasks. Korean patent application pending, 2025.

Skills

Robot learning: reinforcement learning (PPO, multi-agent RL), imitation learning (behavior cloning, DAgger-style online distillation), action chunking (ACT), transformer policies, teacher-student policy distillation, sim-to-real transfer, automatic domain randomization, teleoperation data collection
Currently studying: VLA policies (SmolVLA fine-tuning in progress), Diffusion Policy and flow matching, GAIL and inverse RL
Perception: 6D object pose estimation and tracking (FoundationPose), open-vocabulary detection and segmentation (YOLO-World, SAM 2), RGB visuomotor policies, multi-camera RGB-D (Intel RealSense)
Simulation: NVIDIA Isaac Lab, Isaac Sim, Isaac Gym, digital twins, massively parallel GPU training
Software: Python, C++, PyTorch (incl. DDP), CUDA, ROS 2, Docker, Linux, Git, LeRobot datasets, cuRobo
Hardware: UR3 and UR3e arms, Allegro Hand V4, Manus gloves, VIVE trackers, Siemens PLCs, LiDAR

Open-Source Projects (GitHub)

- **dexterous-pick-and-place:** FoundationPose 6D perception with BODex force-closure grasp synthesis (cuRobo) on a UR3 with an Allegro Hand; 94 of 100 real-robot trials over 10 objects, 180 of 200 in simulation with both arms.
- **dual-arm-allegro-ros2** / **allegro-hand-retargeting:** one ROS 2 Jazzy stack for the two UR arms, two Allegro hands (modified CAN driver) and RealSense cameras; Manus-glove-to-Allegro retargeting (GeoRT), 0.7 ms per hand.
- **FoldingACT (IROS 2026 Robotic Origami Challenge):** vision-tactile ACT policy (DINOv3, Sparsh) trained with PyTorch DDP on a 398 GB LeRobot dataset; the final v10, run by the organizers at the IROS 2026 venue, completed 2 of the 6 folds; Team PixelPaper placed 5th.

Languages

Spanish (native) | English (fluent) | Korean (basic; preparing for TOPIK)